

The Laplace transform in Lean 4

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Abstract

This paper details the rigorous formalization of the Laplace transform and its inverse in the Lean 4 proof assistant. We verify fundamental operational properties—such as linearity, frequency shifting, and the n -th derivative rule—while navigating Lean’s strict measure-theoretic and topological constraints. Crucially, because Mathlib currently lacks extensive complex contour integration, we bypass the classical Bromwich contour to construct a purely real-analytic proof of the inversion theorem. This approach utilizes Fubini’s theorem and requires the formal evaluation of the Dirichlet integral, which we achieve via differentiation under the integral sign and the Dominated Convergence Theorem.

1 Introduction

The Laplace transform [1] is an integral transform, initially utilized by Pierre-Simon Laplace in the context of probability theory, which maps real-valued functions into the complex frequency domain. It serves as a standard method in physics and engineering, particularly due to its utility in solving linear ordinary and partial differential equations.

Classically, the Laplace transform of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined as the following improper integral:

$$\mathcal{L}\{f(t)\}(s) := \int_0^\infty e^{-st} f(t) dt, \quad (1)$$

where $s \in \mathbb{C}$. By converting differential equations in the time domain into algebraic equations in the s -domain, the transform facilitates the analysis of linear time-invariant systems, which is a core component of control theory and signal processing.

Formalizing the Laplace transform in a proof assistant such as Lean 4 presents specific theoretical and computational challenges. Historically, interactive theorem

proving has focused heavily on discrete mathematics, logic, and algebra. Continuous mathematics—involving measure theory, Lebesgue integration, limits at infinity, and complex analysis—requires extensive foundational libraries. By rigorously formalizing the Laplace transform and its inverse, this work contributes to the integration of applied mathematics within verified environments.

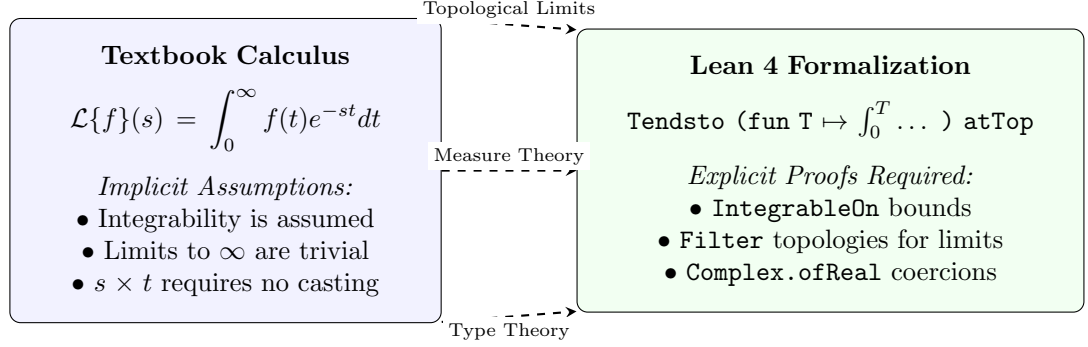


Fig. 1 The Formalization Gap: Translating the implicit assumptions of informal continuous calculus into the rigorous, foundational structures required by the Lean 4 environment.

In this paper, we define the Laplace transform and formally prove its primary properties within the Lean 4 environment.

- In Section 2, we provide the foundational definitions and topological assumptions used in our formalization, progressing from a generalized space to the classical real-to-complex case.
- In Section 3, we derive the elementary properties of the transform and detail the proofs for standard operational rules.
- In Section 4, we define the inverse Laplace transform. Because full complex contour integration is not currently supported in Mathlib, we construct a real-analytic proof of the inversion theorem utilizing the Dirichlet integral.
- In Section 5, we formalize the evaluation of the Dirichlet integral itself. This requires overcoming the non-Lebesgue integrability of the sinc function by employing differentiation under the integral sign and the Dominated Convergence Theorem to rigorously evaluate the necessary limits.

Throughout each section, we discuss the Lean 4 specific challenges—such as managing type coercions, filter topologies, and strict interval integrability conditions—that arise when translating informal analysis into a formal proof environment.

2 Definition of the Laplace Transform

Before analyzing specific functions and their transforms, we must establish a robust foundational definition. In our Lean formalization, we first build a generalized version

of the Laplace transform over arbitrary topological and normed spaces, before specializing it to the classical real-to-complex case that engineers and physicists use in practice.

2.1 Generalized to classical Laplace Transform

We begin by establishing the most general setting possible for the Laplace transform. Classically, the Laplace transform integrates a function over the real line. However, the algebraic essence of the transform—multiplying a function by a decaying exponential kernel and integrating over a measure space—can be abstracted.

We define a generalized space E which is equipped with several algebraic and topological structures. In Lean, we specify that E is a complete normed space over the complex numbers $\mathbb{C}$, equipped with a measurable space structure to allow for integration:

```
1 variable {E : Type*} [NormedRing E] [CompleteSpace E]
2 [NormedSpace ℂ E] [NormedAlgebra ℂ E] [MeasurableSpace E]
3 [IsBoundedSMul ℂ E] [NormedSpace ℝ E] [SMulCommClass ℝ ℂ E]
```

With these structures in place, we define the kernel of the generalized transform. The classical term e^{-st} is abstracted to a generic linear map application evaluated through a normed space exponential.

```
1 def laplaceKernel (S : Set E) (L : S → ℂ → E) (e : S) (p : ℂ) : E :=
2   NormedSpace.exp ℂ (- (L e p))
3
4 def fullLaplaceKernel (S : Set E) (L : S → ℂ → E) (f : S → E) (p : ℂ) : S → E :=
5   fun e ↦ f e * (laplaceKernel S L e p) • (1 : E)
```

The `laplaceKernel` computes the abstract exponential $e^{-L(e,p)}$, and the `fullLaplaceKernel` multiplies this exponentially decaying term by the function $f(e)$ that we are transforming. With the integrand fully defined, the generalized Laplace transform is simply the Bochner integral of this kernel over a measure space S equipped with a measure μ :

```
1 def GeneralizedLaplaceTransform (S : Set E) (L : S → ℂ → E) (f : S → E) (μ : Measure S) : ℂ → E
2   :=
3   fun p ↦ ∫ e, fullLaplaceKernel S L f p e ∂μ
```

While this generalized form works well for abstract algebraic proofs (such as the theorems `GeneralizedLaplaceTransform.const.smul` and `GeneralizedLaplaceTransform.additive` which establish base linearity), our primary goal is the classical Laplace transform mapping $f : \mathbb{R} \rightarrow \mathbb{C}$.

To map this abstraction to the classical case, we restrict our generalized space to the real line, embedded in the complex plane. We define `realLine` as the set of complex numbers where the imaginary part is zero, and we define our standard Lebesgue measure restricted to the non-negative reals $[0, \infty)$:

```
1 def non_negative_reals : Set ℝ := Ici 0
2
3 def μ_real : Measure ℝ := volume.restrict non_negative_reals
4 def μ_r : Measure realLine := Measure.map real_to_realLine μ_real
```

This allows us to write down the classical Laplace transform precisely as it appears in textbooks. However, because dealing with limits at infinity is computationally tricky in formal proofs, we split the definition into two parts. First, we define the standard Laplace transform on $(0, \infty)$:

```
1 noncomputable def laplaceTransform (f : ℝ → ℂ) (s : ℂ) : ℂ :=
2   ∫ t in Ioi 0, f t * cexp (-s * t)
```

Second, we define the *finite* or *truncated* Laplace transform, which integrates only up to a finite time T :

```
1 noncomputable def finiteLaplaceTransform (f : ℝ → ℂ) (s : ℂ) (T : ℝ) : ℂ :=
2   ∫ t in Ioc 0 T, f t * cexp (-s * t)
```

The bridge between these two definitions is formalized by the theorem `finite_laplace_tendsto_laplace`. It states that if the integrand is Lebesgue-integrable on $(0, \infty)$, the limit of the finite transform as $T \rightarrow \infty$ is exactly the infinite Laplace transform:

$$\lim_{T \rightarrow \infty} \int_0^T f(t) e^{-st} dt = \int_0^\infty f(t) e^{-st} dt \quad (2)$$

2.2 Challenges

Defining the transform in Lean introduces several hurdles not present in informal mathematics.

The first major challenge lies in **Lebesgue Integration and Integrability**. In Mathlib, the integral of a function that is not strictly integrable evaluates to 0 by default. This “junk value” prevents us from carelessly writing down transforms. Every time we claim a property about the Laplace transform, we must explicitly pass a proof of integrability (e.g., `IntegrableOn (fun t ↦ f t * cexp (-s * t)) (Ioi 0)`).

The second challenge is the **Dominated Convergence Theorem (DCT)**. To prove that the finite transform converges to the infinite transform, we cannot just swap the limit and the integral. We had to construct an indicator function `(Iic T).indicator` that evaluates to 1 if $t \leq T$ and 0 otherwise. We then applied the filter limit `tendsto_integral_filter_of_dominated_convergence`, proving that the absolute norm $\|f(a)\| \cdot e^{-\Re(s)a}$ serves as a valid dominating function independent of T .

Finally, **Type Coercion** between $\mathbb{R}$ and $\mathbb{C}$ creates immense friction. The variable t is a real measure of time, but the frequency s and the result are complex. The expression $-s * t$ requires casting t to $\mathbb{C}$. In Lean, managing these casts (e.g., `push_cast`, `Complex.ofReal_re`) demands extensive algebraic manipulation to prove that seemingly trivial equalities like $\Re(-s \cdot t) = -\Re(s) \cdot t$ hold true.

3 Elementary properties of the Laplace transform

With the foundational definitions established and the equivalence between the limit of the finite transform and the global transform proven, we can now derive the classical properties and elementary pairs of the Laplace transform.

3.1 Proofs and Formalized Properties of the Laplace Transform

The elementary properties of the Laplace transform generally fall into two categories: operational properties (like linearity and the derivative rules) and explicit transform evaluations (like the transforms of 1, t , and e^{at}). While evaluating these transforms informally in classical calculus is often a matter of trivial algebraic manipulation and basic limit evaluation, formalizing them in Lean 4 requires decomposing every assumption into rigorous topological, measure-theoretic, and analytic steps.

To summarize our results, Table 1 presents the mathematical properties successfully formalized in our Lean 4 project alongside their corresponding theorem names.

Property / Function	Lean 4 Theorem Name	Mathematical Result
Generalized Additivity	<code>GeneralizedLaplaceTransform.additive</code>	$\mathcal{L}\{f_1 + f_2\} = \mathcal{L}\{f_1\} + \mathcal{L}\{f_2\}$
Generalized Scalar Mul	<code>GeneralizedLaplaceTransform.const.smul</code>	$\mathcal{L}\{r \cdot f\} = r \cdot \mathcal{L}\{f\}$
Generalized Freq. Shift	<code>GeneralizedLaplaceTransform.complex.add</code>	$\mathcal{L}\{f\}(p_1 + p_2) = \mathcal{L}\{\mathcal{L}_{\text{freq}}(f, p_1)\}(p_2)$
Finite Linearity	<code>finiteLaplaceTransform.linear</code>	$\mathcal{L}_T\{af + bg\}(s) = a\mathcal{L}_T\{f\} + b\mathcal{L}_T\{g\}$
Constant: $f(t) = 1$	<code>laplace_of_one_eq_inv_s</code>	$\mathcal{L}\{1\}(s) = \frac{1}{s}$
Linear: $f(t) = t$	<code>laplace_of_t_eq_inv_s_sq</code>	$\mathcal{L}\{t\}(s) = \frac{1}{s^2}$
Power: $f(t) = \frac{t^k}{k!}$	<code>finiteLaplaceTransform.pow_div_factorial</code>	$\mathcal{L}_T\left\{\frac{t^k}{k!}\right\} = \frac{1}{s^{k+1}} \left(1 - e^{-sT} \sum_{j=0}^k \frac{(sT)^j}{j!}\right)$
Exponential: $f(t) = e^{at}$	<code>finite_laplace_of_exp_eq</code>	$\mathcal{L}_T\{e^{at}\}(s) = \frac{1 - e^{-(s-a)T}}{s-a}$
First Derivative: $f'(t)$	<code>finite_laplace_deriv_eq</code>	$\mathcal{L}_T\{f'\}(s) = s\mathcal{L}_T\{f\} - f(0) + e^{-sT}f(T)$
n -th Derivative: $f^{(n)}(t)$	<code>finite_laplace_iteratedDeriv_eq</code>	$\mathcal{L}_T\{f^{(n)}\}(s) = s^n \mathcal{L}_T\{f\} - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0) + \dots$

Table 1 Summary of Laplace transform properties and basic pairs formalized in Lean 4.

Below, we detail the formalization of the fundamental operational properties and the base transform pairs.

3.1.1 Linearity of the Finite Transform

For the finite (truncated) Laplace transform over the interval $(0, T]$, we formally establish linearity via the theorem `finiteLaplaceTransform.linear`:

$$\mathcal{L}_T\{af(t) + bg(t)\}(s) = a\mathcal{L}_T\{f(t)\}(s) + b\mathcal{L}_T\{g(t)\}(s) \quad (3)$$

In Lean 4, this is stated by explicitly passing integrability hypotheses for both functions:

```

1 theorem finiteLaplaceTransform_linear
2   (f g : ℝ → ℂ) (s a b : ℂ) (T : ℝ)
3   (hf : IntegrableOn (fun t ↦ f t * cexp (-s * t)) (Ioc 0 T))
4   (hg : IntegrableOn (fun t ↦ g t * cexp (-s * t)) (Ioc 0 T)) :
5   finiteLaplaceTransform (fun t ↦ a * f t + b * g t) s T =
6   a * finiteLaplaceTransform f s T + b * finiteLaplaceTransform g s T

```

This relies on Mathlib’s underlying theorems for Lebesgue integration, specifically `integral.add` and `integral.const_mul`. However, Lean’s integration API strictly

demands proof of integrability before allowing an integral to be split. Therefore, the theorem explicitly requires the hypotheses `hf` and `hg`, which state `IntegrableOn (fun t ↦ f t * cexp (-s * t)) (Ioc 0 T)` and identically for `g`. Without these proofs passed as arguments, the integrals evaluate to 0 by convention, and the equality fails.

3.1.2 The Constant Function ($f(t) = 1$)

The simplest explicit transform pair is the transform of the constant function 1. We tackle this in two phases: the finite evaluation and the global limit.

1. The Finite Transform:

We formalize the truncated transform in `finite_laplace_of_one_eq_eval`:

$$\mathcal{L}_T\{1\}(s) = \frac{e^{-sT}}{-s} - \frac{e^{-s0}}{-s} \quad (4)$$

The Lean formulation ensures that division by zero is avoided:

```
1 theorem finite_laplace_of_one_eq_eval (s : ℂ) (T : ℝ) (hs : s ≠ 0) (hT : 0 ≤ T) :
2   finiteLaplaceTransform (fun _ ↦ 1) s T = (cexp (-s * T) / -s) - (cexp (-s * 0) / -s)
```

To prove this, we define the exact antiderivative $F(t) = e^{-st} / -s$. Applying the Fundamental Theorem of Calculus in Lean (`intervalIntegral.integral_eq_sub_of_hasDerivAt`) is quite strict. We must provide a pointwise proof that $F'(t) = e^{-st}$ for all t in the closed interval. This is achieved by mapping the chain rule via `HasDerivAt.ofReal_comp`, `hasDerivAt_id`, and `HasDerivAt.div_const`. Furthermore, we must prove the resulting integrand is continuous (and therefore interval integrable) using the `continuity` tactic. The proof requires the explicit hypothesis that $s \neq 0$ (`hs : s ≠ 0`) to satisfy Lean's field simplification rules (`field_simp`).

2. The Global Limit:

To transition to the standard infinite Laplace transform, we prove `laplace_of_one_eq_inv_s`:

$$\mathcal{L}\{1\}(s) = \frac{1}{s} \quad (5)$$

```
1 theorem laplace_of_one_eq_inv_s (s : ℂ) (hs : 0 < s.re) :
2   laplaceTransform (fun _ ↦ 1) s = 1 / s
```

This theorem requires the strict decay condition $\Re(s) > 0$. In Lean, dealing with limits at infinity requires the `Filter` topology. We prove that the sequence of finite transforms converges to $1/s$ using `tendsto_nhds_unique`. We decouple the finite evaluation into $(1/-s) \cdot e^{-sT} + (1/s)$ and prove that the exponential term vanishes as $T \rightarrow \infty$. This is done by casting the complex norm into a real exponential `Real.exp(-s.re * T)` and applying `Real.tendsto_exp_atBot` alongside filter arithmetic (`Filter.Tendsto.add`, `Filter.Tendsto.const_mul`).

3.1.3 The Linear Function ($f(t) = t$)

Evaluating the transform of t introduces the necessity of Integration by Parts (IBP).

1. The Finite Transform:

In `finite_laplace_of_t_eq_eval`, we prove:

$$\mathcal{L}_T\{t\}(s) = \frac{1}{s^2}(1 - e^{-sT}) - \frac{Te^{-sT}}{s} \quad (6)$$

```
1 theorem finite_laplace_of_t_eq_eval (s : ℂ) (T : ℝ) (hs : s ≠ 0) (hT : 0 ≤ T) :
2   finiteLaplaceTransform (fun t ↦ (t : ℂ)) s T =
3   (1 / s^2) * (1 - cexp (-s * T)) - ((T : ℂ) * cexp (-s * T) / s)
```

The Lean proof utilizes `intervalIntegral.integral_mul_deriv_eq_deriv_mul`. We define $u(t) = t$ and $v(t) = e^{-st}/-s$. We must independently prove `HasDerivAt` for both functions, as well as their interval integrability over the compact interval $[0, T]$. Once IBP is applied, the proof state yields a new integral $\int_0^T e^{-st}/-s dt$. We must then apply the Fundamental Theorem of Calculus a second time, defining a new antiderivative $F(t) = e^{-st}/(-s)^2$, differentiating it, and evaluating the boundaries to close the proof via algebraic ring simplification.

2. The Global Limit:

In `laplace_of_t_eq_inv_s_sq`, we prove the global convergence $\mathcal{L}\{t\}(s) = 1/s^2$.

```
1 theorem laplace_of_t_eq_inv_s_sq (s : ℂ) (hs : 0 < s.re) :
2   laplaceTransform (fun t ↦ t) s = 1 / s^2
```

Similar to the constant function, this requires $\Re(s) > 0$. The primary challenge here is proving that the boundary term Te^{-sT} converges to 0 as $T \rightarrow \infty$. This requires combining polynomial growth with exponential decay within Lean's filter limits, manipulating the complex norms to rely on the real-valued limit properties of asymptotic decay.

3.1.4 The Exponential Function ($f(t) = e^{at}$)

A core component of analyzing linear systems is the frequency-shifting property, modeled by the transform of the exponential function. We prove the finite evaluation in `finite_laplace_of_exp_eq`:

$$\mathcal{L}_T\{e^{at}\}(s) = \frac{1 - e^{-(s-a)T}}{s - a} \quad (7)$$

```
1 theorem finite_laplace_of_exp_eq (a s : ℂ) (T : ℝ) (h : s ≠ a) (hT : 0 ≤ T) :
2   finiteLaplaceTransform (fun t ↦ cexp (a * t)) s T = (1 - cexp (-(s - a) * T)) / (s - a)
```

We combine the exponents $e^{at}e^{-st} = e^{-(s-a)t}$. In Lean, this requires explicitly rewriting the terms using `Complex.exp_add` and manipulating the algebraic ring (`sub_eq_add_neg`).

Once combined, we define the specific antiderivative $F(t) = e^{(a-s)t}/(a-s)$. Because the denominator contains a complex subtraction, the Fundamental Theorem of Calculus evaluation requires a strict hypothesis that $s \neq a$ (`h : s ≠ a`), allowing us to use `sub_ne_zero.mpr` to prove the denominator is non-zero during the derivative check.

The remainder of the proof consists of evaluating the bounds at T and 0 , simplifying $e^0 = 1$, and carefully distributing the negative signs (`neg_sub`) to arrive at the exact analytical form.

3.2 Formalizing Complex Properties via Mathematical Induction

While simple properties rely strictly on evaluating antiderivatives, more complex formulas—such as the power rule and the general derivative rule—require formal mathematical induction. Translating induction from informal mathematics into a proof assistant involves rigorously managing boundaries, maintaining analytical assumptions across inductive steps, and manipulating summation indices using Lean’s `Finset` API.

3.2.1 Transform of the Power Function ($t^k/k!$)

To compute the Laplace transform of the scaled power function $f(t) = \frac{t^k}{k!}$, we formalize the theorem `finiteLaplaceTransform_pow_div_factorial` by inducting on $k \in \mathbb{N}$. The mathematical goal is to prove the truncated expression:

$$\mathcal{L}_T \left\{ \frac{t^k}{k!} \right\} (s) = \frac{1}{s^{k+1}} \left(1 - e^{-sT} \sum_{j=0}^k \frac{(sT)^j}{j!} \right) \quad (8)$$

```
1 theorem finiteLaplaceTransform_pow_div_factorial (k : ℕ) (s : ℂ) (T : ℝ) (hs : s ≠ 0) (hT : 0 ≤ T)
2   :
3   finiteLaplaceTransform (fun t => (t : ℂ) ^ k / (k.factorial : ℂ)) s T =
  (1 / s^(k + 1)) * (1 - cexp (-s * (T : ℂ))) * ∑ j ∈ Finset.range (k + 1), (s * (T : ℂ))^j / (j
    .factorial : ℂ))
```

1. The Base Case:

The proof begins with the base case (`case zero`), where $k = 0$. Because $0! = 1$, the function algebraically reduces to $f(t) = 1$. As established in our elementary properties, evaluating this via the Fundamental Theorem of Calculus (FTC) yields $\frac{e^{-sT}}{-s} - \frac{1}{-s}$. In Lean, we use the `ring` and `field.simp` tactics to demonstrate that this resulting expression simplifies exactly to the target formula when evaluated at $k = 0$.

2. The Inductive Step:

The inductive step (`case succ k ih`) assumes the formula holds for k and attempts to prove it for $k + 1$. This is executed via a rigorous application of Integration by Parts (IBP). We define our component functions and explicitly provide their Lean derivatives:

$$u(t) = \frac{t^{k+1}}{(k+1)!} \implies u'(t) = \frac{t^k}{k!} \quad (9)$$

$$v(t) = \frac{e^{-st}}{-s} \implies v'(t) = e^{-st} \quad (10)$$

Proving $u'(t)$ in Lean introduces significant type-casting friction. Because k is a natural number ($\mathbb{N}$), we must use `Nat.factorial_succ` and cast the arithmetic into the complex field ($\mathbb{C}$) using `Nat.cast_mul` and the `push_cast` tactic to formally prove that $(k+1)! = (k+1) \cdot k!$.

After providing pointwise proofs of differentiability (`HasDerivAt`), applying `intervalIntegral.integral_mul_deriv_eq_deriv_mul` (the IBP theorem) trades our original integral for the evaluated boundary terms minus a new integral:

$$\int_0^T \frac{t^{k+1}}{(k+1)!} e^{-st} dt = \left[\frac{t^{k+1}}{(k+1)!} \frac{e^{-st}}{-s} \right]_0^T - \int_0^T \frac{t^k}{k!} \frac{e^{-st}}{-s} dt \quad (11)$$

Evaluating the boundaries in Lean requires applying the `zero_pow` lemma to prove the lower bound vanishes, leaving only the T term. We then extract the constant $\frac{1}{s}$ from the remaining integral, recognizing that what remains is precisely the definition of the truncated Laplace transform for k .

At this point, we apply our inductive hypothesis (`ih`). The remaining challenge is algebraic: absorbing the boundary term into the `Finset` summation provided by the inductive hypothesis. We use the `Finset.sum_range_succ` theorem multiple times to shift the indices, proving that the isolated boundary term represents the exact $j = k+1$ term of the expanded summation, successfully closing the induction.

3.2.2 Transform of the n -th Derivative ($f^{(n)}(t)$)

One of the most used properties of the Laplace transform is its ability to algebraicize differential equations. We first prove the base derivative theorem `finite_laplace_deriv_eq`:

$$\mathcal{L}_T\{f'(t)\}(s) = s\mathcal{L}_T\{f(t)\}(s) - f(0) + e^{-sT}f(T) \quad (12)$$

```
1 theorem finite_laplace_deriv_eq (f df : ℝ → ℂ) (s : ℂ) (T : ℝ)
2   (hT : 0 ≤ T)
3   (hf : ∀ t ∈ Set.Icc 0 T, HasDerivAt f (df t) t)
4   (hf_int : IntervalIntegrable df volume 0 T)
5   (h_cont : ContinuousOn df (Set.Icc 0 T))
6   (hf_int : IntervalIntegrable df volume 0 T) :
7   finiteLaplaceTransform df s T = s * finiteLaplaceTransform f s T - f 0 + cexp (-s * T) * f T
```

This is proven via direct Integration by Parts where $u(t) = e^{-st}$ and $v'(t) = f'(t)$. To extend this to the n -th derivative, we state `finite_laplace_iteratedDeriv_eq` and induct on $n \in \mathbb{N}$:

$$\mathcal{L}_T\{f^{(n)}\} = s^n \mathcal{L}_T\{f\} - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0) + e^{-sT} \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(T) \quad (13)$$

```
1 theorem finite_laplace_iteratedDeriv_eq (f : ℝ → ℂ) (s : ℂ) (T : ℝ) (n : ℕ)
2   (hT : 0 ≤ T)
3   (hf_cont_diff : ContDiff ℝ (n) f)
4   (hf_int : IntervalIntegrable (iteratedDeriv n f) volume 0 T) :
```

```

5  finiteLaplaceTransform (iteratedDeriv n f) s T =
6    s^n * finiteLaplaceTransform f s T
7    - (∑ k ∈ Finset.range n, s^(n - 1 - k) * iteratedDeriv k f 0)
8    + cexp (-s * T) * (∑ k ∈ Finset.range n, s^(n - 1 - k) * iteratedDeriv k f T)

```

1. The Base Case:

For $n = 0$, the zero-th derivative is simply f , and the summations over an empty range (`Finset.range 0`) automatically evaluate to zero. This trivially yields the identity $\mathcal{L}_T\{f\} = \mathcal{L}_T\{f\}$.

2. The Inductive Step:

For $n + 1$, we treat the $(n + 1)$ -th derivative as the first derivative of $f^{(n)}$ and directly apply our previously proven first-derivative rule. However, Lean’s rigorous type system requires us to prove that $f^{(n)}$ is actually differentiable. We achieve this by passing a hypothesis of continuous differentiability (`ContDiff ℝ n f`) and using Mathlib’s `ContDiff.continuous.iteratedDeriv` to prove that the derivatives are continuous, which in turn guarantees they are `IntervalIntegrable`.

Substituting the inductive hypothesis yields a massive algebraic expression. We must distribute the s multiplier into both summations using `Finset.mul_sum`. Lean requires explicit index arithmetic to verify that $s \cdot s^{n-1-k} = s^{n-k}$, a step we easily manage using the `omega` tactic (designed for linear integer arithmetic). Finally, we apply `Finset.sum_range_succ` in reverse to fold the isolated boundary terms, $-f^{(n)}(0)$ and $e^{-sT}f^{(n)}(T)$, back into the summation. This formally extends the upper bound of the sum from n to $n + 1$, completing the algebraic equivalence.

3.2.3 Challenges in Formalization

Proving these elementary properties highlighted the strict rigor required when working with continuous calculus in Lean 4.

- **Strictness of the Fundamental Theorem of Calculus:** In informal mathematics, substituting the boundaries of an antiderivative is trivial. In Lean, applying `integral.eq_sub_of_hasDerivAt` strictly demands three things: the exact definition of the antiderivative function F , a pointwise proof of differentiability (`HasDerivAt`) everywhere on the closed interval, and a proof that the resulting integrand is continuous. Every single constant, such as $\frac{1}{-s}$, requires explicitly mapping the chain rule via `HasDerivAt.comp` and `HasDerivAt.const_mul`.
- **Interval Integrals vs. Set Integrals:** Mathlib makes a distinct topological separation between integrating over a set (like $\int_{x \in A}$) and integrating over an oriented interval ($\int_a^b$). The finite Laplace transform is defined using the set integral `Ioc 0 T`. However, integration theorems like IBP only apply to `intervalIntegral`. We constantly had to rewrite our integrals using `intervalIntegral.integral_of_lehT` to bounce between the two formalisms, requiring continuous, explicit proofs that $0 \leq T$.
- **Limits and Filters:** Taking limits as $T \rightarrow \infty$ requires reasoning in Mathlib’s `Filter` topology. We cannot simply state that $e^{-sT} \rightarrow 0$. Instead, we must prove that the norm of the complex exponential evaluates to $e^{-\Re(s)T}$, and then use

`Filter.Tendsto` along the `atTop` filter to prove it converges to the `nhds 0` (neighborhood of zero). Constructing these filter limits requires careful algebraic tracking of the real and imaginary components of s to ensure the decay condition $\Re(s) > 0$ is strictly respected.

4 Inverse Laplace Transform and Dirichlet Integral

4.1 A presentation of the inverse and an example of application

An interesting property of the Laplace Transform is the existence of its inverse. In other words, there is an operator $\mathcal{L}^{-1}$ such that $\mathcal{L}^{-1}(\mathcal{L}(f)) = f$, provided that the function $f(t)$ satisfies specific analytical conditions. By Lerch's theorem, this inverse is unique (almost everywhere) as long as $f(t)$ exhibits the following properties:

1. Causality: The function is equal to zero for $t < 0$.
2. Piecewise Continuity: The function $f(t)$ is piecewise continuous on the interval $[0, \infty)$.
3. Exponential Order: The function does not grow faster than an exponential. There must exist real constants $M > 0$ and c such that $|f(t)| \leq Me^{ct}$ for all $t \geq 0$.

When these conditions are met, the inverse Laplace transform can be explicitly computed using Mellin's inverse formula, also known as the Bromwich integral. If $F(s) = \mathcal{L}(f)(s)$, the original function $f(t)$ is recovered via a contour integration in the complex plane:

$$f(t) = \frac{1}{2\pi i} \lim_{T \rightarrow \infty} \int_{\gamma - iT}^{\gamma + iT} F(s) e^{st} ds$$

Here, s is the complex frequency parameter ($s = \sigma + i\omega$), and γ is a real constant chosen such that it is greater than the real parts of all singularities of $F(s)$. This ensures that the vertical integration path lies entirely within the region of absolute convergence of the transform.

The interest of the inverse Laplace transform becomes blatant when solving differential equations. Indeed, the properties of the Laplace transform (e.g the derivative) would be strictly theoretical without the ability to map the algebraic solution back to the time domain.

4.2 Lean implementation of Inverse Laplace Transform, some properties and Implementation issues

In this section, we present the construction of the inverse operator, some algebraic properties we proved in our Lean code, and present the classical inversion approach using residue theory. Finally, we discuss the technical hurdles that prevent a direct implementation of this classical complex-analytic proof in the current state of Lean's Mathlib.

4.2.1 Defining the Inverse Laplace Transform in Lean

The Bromwich integral, presented earlier, is performed along a vertical contour in the complex plane, parametrized by $s = \gamma + iT$, where γ is a fixed real number and T

varies across the real line. To implement this in Lean, our first step is to abstract this parametrization. We define `imNbFromReals` to construct this complex number from two real variables:

```
1 def imNbFromReals (γ : ℝ) (T : ℝ) : ℂ :=
2   (γ : ℂ) + (T : ℂ) * I
```

The core of the inverse operator lies in the integrand, which we define as `InverseLaplaceKernel`. By currying the arguments and defining `InverseLaplaceKernelFunctT`, we isolate the variable T to prepare the function for Lebesgue integration over $\mathbb{R}$:

```
1 def InverseLaplaceKernel (F : ℂ → ℂ) (t : ℝ) : ℝ → ℝ → ℂ :=
2   fun γ T ↦ I*(Complex.exp ((imNbFromReals γ T) * t)) * F (imNbFromReals γ T)
3
4 def InverseLaplaceKernelFunctT (F : ℂ → ℂ) (t : ℝ)(γ : ℝ) : ℝ → ℂ :=
5   (InverseLaplaceKernel F t) γ
```

The inverse transform itself is formally expressed via the `inverseLaplace_t` definition, which computes the integral:

$$f(t) = \frac{1}{2\pi i} \int_{-\infty}^{\infty} i e^{(\gamma + iT)t} F(\gamma + iT) dT$$

Notice that the factors of i are preserved to maintain the distinction between the universal Cauchy constant $\frac{1}{2\pi i}$ and the change of measure $ds = i dT$ arising from the integration over the vertical line $\Re(s) = \gamma$.

Writing down an integral in Lean is not sufficient to guarantee its mathematical validity. To ensure the Lebesgue integral does not silently evaluate to zero, we need to provide a proof of integrability (`h_integrable_in_S`) over a set of times S as an explicit argument.

```
1 def inverseLaplace_t (F : ℂ → ℂ) (γ t : ℝ) : ℂ :=
2   1/(2*I*Real.pi) * ∫ T : ℝ, InverseLaplaceKernel F t γ T
3
4 def inverseLaplaceFunction (F : ℂ → ℂ) (γ : ℝ) (S : Set ℝ)
5   (h_integrable_in_S : ∀ t ∈ S, Integrable ((InverseLaplaceKernelFunctT F t) γ) volume)
6   : S → ℂ :=
7   fun t ↦ inverseLaplace_t F γ t
```

4.2.2 Basic Properties: Linearity of the Operator

A fundamental requirement for the inverse Laplace transform is linearity, which is critical for solving linear differential equations. In our formalization, we prove that our Lean operator respects scalar multiplication and additivity.

- **Additivity (`inverseLaplaceAdditive_t`):** This theorem establishes that the inverse transform of a sum is the sum of the inverse transforms. The proof is structured in two steps: first, we establish additivity at the kernel level. As shown in the Lean snippet below, we rely on function extensionality (`funext`) and the distributive property of complex numbers to decompose the integrand:

```
1 theorem InverseLaplaceKernelAdditive (F1 F2 : ℂ → ℂ) (t : ℝ) :
```

```

2 InverseLaplaceKernel (F1 + F2) t = InverseLaplaceKernel F1 t + InverseLaplaceKernel F2 t := by
3   funext γ T
4   unfold InverseLaplaceKernel
5   calc I * cexp (imNbFromReals γ T * ↑t) * (F1 + F2) (imNbFromReals γ T)
6   _ = I * cexp (imNbFromReals γ T * ↑t) * (F1 (imNbFromReals γ T) + (imNbFromReals γ T))
7   := by
8     simp_all only [Pi.add_apply]
9     _ = I * cexp (imNbFromReals γ T * ↑t) * F1 (imNbFromReals γ T) +
10       I * cexp (imNbFromReals γ T * ↑t) * F2 (imNbFromReals γ T) := by
11       rw [←left_distrib]
12

```

Once the kernel's additivity is proven, the global additivity of the operator follows directly from the linearity of the integral (using `integral_add`), provided that the functions F_1 and F_2 satisfy the required integrability conditions.

```

1 theorem inverseLaplaceAdditive_t(F1: ℂ → ℂ) (F2: ℂ → ℂ)(γ t : ℝ)
2 (h1 : Integrable (InverseLaplaceKernelFunctT F1 t γ) volume)
3 (h2 : Integrable (InverseLaplaceKernelFunctT F2 t γ) volume):
4   inverseLaplace_t (F1+F2) γ t = inverseLaplace_t F1 γ t + inverseLaplace_t F2 γ t
5

```

- **Scalar Multiplication (`inverseLaplaceConst_t`):** Similarly, we guarantee that factoring out a complex scalar c is valid. The Lean proof manipulates the complex product to factor c out of the exponential term.

```

1 theorem InverseLaplaceKernelConst (F : ℂ → ℂ) (c : ℂ) (t : ℝ) :
2   InverseLaplaceKernel (c • F) t = c • (InverseLaplaceKernel F t)
3
4 theorem inverseLaplaceConst_t(F: ℂ → ℂ) (c:ℂ)(γ t : ℝ)
5 (h_integrable : Integrable (InverseLaplaceKernelFunctT F t γ) volume) :
6   inverseLaplace_t (c • F) γ t = c * inverseLaplace_t F γ t

```

4.2.3 The Classical Approach: Inversion via Residue Calculus

We wish now to prove that the Bromwich integral is indeed the Laplace Transform inverse. To evaluate the Bromwich integral explicitly, mathematicians usually rely on Complex Analysis, specifically Cauchy's Residue Theorem.

To understand this approach without needing advanced complex analysis, we can introduce the concept of a *pole*. In many practical physical systems, the function $F(s)$ is a fraction (e.g., $F(s) = \frac{1}{s^2+1}$). The values of s where the denominator becomes exactly zero—causing the function to "blow up" to infinity—are called poles. Associated with each pole is a *residue*, a specific number that captures the behavior of the function around that singularity.

If we zoom in on a function near a simple pole s_k , the function locally behaves like $\frac{R}{s-s_k}$. This constant numerator R is the *residue*. Formally, it is the coefficient of the $(s-s_k)^{-1}$ term in the Laurent series expansion of the function around that singularity.

The classical inversion theorem links the integral to these residues:

Theorem (Inversion via Residue Calculus):

Suppose $F(s)$ is a well-behaved function that tends to 0 as s grows large, and has a finite number of poles $s_1, s_2, \dots, s_n$. If we choose our vertical integration line $\Re(s) = \gamma$ such that all these poles lie strictly to its left, then for $t > 0$, the inverse Laplace

transform is simply the sum of the residues of $F(s)e^{st}$ at these poles:

$$f(t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} F(s)e^{st} ds = \sum_{k=1}^n \text{Res}(F(s)e^{st}, s_k)$$

Classical Proof Idea:

The standard proof relies on drawing a closed "D-shaped" contour in the complex plane. This contour consists of the vertical line segment from $\gamma - iR$ to $\gamma + iR$, closed by a large semicircle C_R extending into the left half-plane.

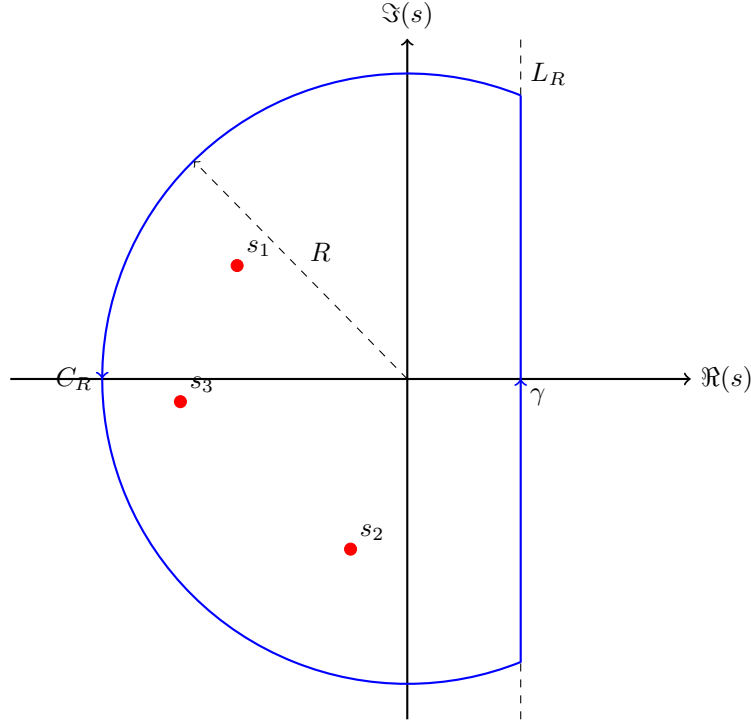


Fig. 2 The closed "D-shaped" Bromwich contour in the complex plane, enclosing the poles s_k of the function.

Cauchy's Residue Theorem states that the integral around this closed loop is proportional to the sum of the enclosed residues. By letting the radius R grow to infinity, the vertical segment becomes the Bromwich integral. Meanwhile, thanks to a result known as *Jordan's Lemma*, the damping effect of e^{st} ensures that the integral along the outer semicircle C_R shrinks to zero. Thus, the Bromwich integral equals the sum of the residues.

4.2.4 Implementation Choices and Mathlib Context

While this geometric approach is standard in university courses, formalizing it directly in Lean requires specific infrastructure. Mathlib’s complex analysis library is steadily growing, and core results like Cauchy’s integral formula are already available. However, dealing with parameterized open contours and geometric limits at infinity involves substantial foundational work.

Specifically, a direct formal proof would require:

1. Defining the precise geometry of the Bromwich contour (combining a line segment with a semicircle).
2. Formalizing Jordan’s Lemma to strictly bound the integral over the expanding arc.
3. Managing the topological proofs required to show that the finite set of poles remains continuously enclosed as the contour grows.

Developing these geometric and complex-measure tools would be a significant project in itself. Therefore, to achieve our goal of formally proving the inverse Laplace transform with the tools currently available in Mathlib, we opt for a purely real-analytic approach. This architectural choice naturally leads us to the Dirichlet integral, which we will explore and implement in the following section.

4.3 Another Strategy: A Real-Analytic Proof

To circumvent the absence of a robust contour integration framework in Mathlib, we decide to prove our Inversion theorem using real-analytic tools. This approach is essentially the Fourier inversion formula in disguise, relying heavily on the Dirichlet integral and integration by parts (see [2]).

4.3.1 The Proof Strategy and the Emergence of the Dirichlet Kernel

We begin directly with the truncated Bromwich integral along the vertical line $s = \gamma + ir$, where r ranges from $-T$ to T :

$$I_T = \frac{1}{2\pi i} \int_{\gamma-iT}^{\gamma+iT} F(s) e^{st} ds = \frac{1}{2\pi} \int_{-T}^T F(\gamma + ir) e^{(\gamma+ir)t} dr$$

At this stage, we replace $F(\gamma + ir)$ with its definition, $F(s) = \int_0^\infty f(a) e^{-sa} da$. This substitutes the complex function with a real integral over a dummy time variable a :

$$I_T = \frac{1}{2\pi} \int_{-T}^T \left(\int_0^\infty f(a) e^{-(\gamma+ir)a} da \right) e^{(\gamma+ir)t} dr$$

This is the pivotal moment of the proof. We have transitioned from a single integral over a complex segment to a double integral over $\mathbb{R}^2$ (the variables a and r). **Assuming absolute convergence**, we apply **Fubini’s theorem** to interchange the order of

integration, grouping the r terms together:

$$I_T = \int_0^\infty f(a) e^{-\gamma(a-t)} \left(\frac{1}{2\pi} \int_{-T}^T e^{ir(t-a)} dr \right) da$$

Evaluating the inner integral with respect to r yields the following expression:

$$I_T = \int_0^\infty f(a) e^{-\gamma(a-t)} \frac{\sin(T(t-a))}{\pi(t-a)} da$$

At this stage, a specific mathematical phenomenon occurs. The term $\frac{\sin(T(t-a))}{\pi(t-a)}$ is a highly oscillatory function that "concentrates" its mass around the point $a = t$ as T grows. To prove that this integral actually recovers the value $f(t)$, we must show that the oscillations everywhere else cancel each other out. This requires the evaluation of the classical Dirichlet integral:

$$\int_0^\infty \frac{\sin(x)}{x} dx = \frac{\pi}{2}$$

This result is the key that allows the integral to act as a "filter" that extracts $f(t)$ from the envelope.

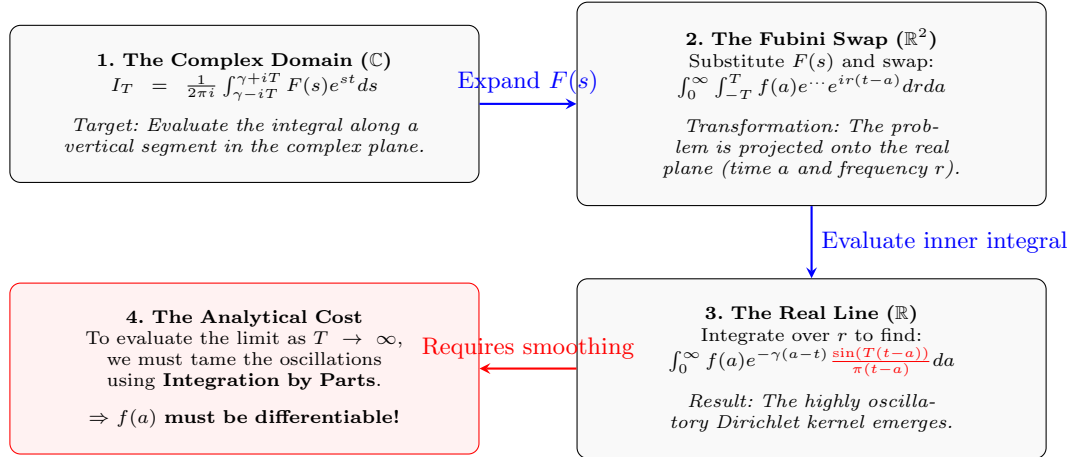


Fig. 3 The structure of the real-analytic proof.

4.3.2 The Shift in Hypotheses

The transition from complex contours to a real-analytic proof shifts the focus from global geometry to local regularity.

The classical approach is primarily **topological**. By using the residue theorem on a closed Bromwich contour, it bypasses the internal oscillations of the integrand. The requirements on $f(t)$ are purely external: piecewise continuity and exponential growth bounds are sufficient to ensure the integral over the large arc vanishes.

Conversely, our formalization evaluates the limit of the integral directly on the real line. To rigorously handle the term $\frac{\sin(T(t-a))}{\pi(t-a)}$ as $T \rightarrow \infty$, we utilize **integration by parts** to shift the limit from the oscillatory kernel to its primitive (the Dirichlet sine function). This identity transfers the regularity requirement from the function f to its derivative f' , leading to stricter analytical constraints:

1. **Global Differentiability:** The function $f(t)$ must be differentiable on $\mathbb{R}$ to satisfy the requirements of the integration by parts formula.
2. **Derivative Integrability:** The term $f'(t)e^{-\gamma t}$ must be Lebesgue-integrable to permit the application of the Dominated Convergence Theorem during the final limiting process.

4.3.3 Assembling the Final Inversion Theorem

With the Dirichlet kernel formalized, we construct the global inversion theorem. The integration into our main project is done in several distinct steps to satisfy Lean's rigorous type checker:

First, we prove Fubini's theorem on the truncated interval $[-T, T]$ (`Fubini.lemma`), and simplify the complex exponentials into real sine functions (`integrand_simplification`). This translates the complex integral into the real domain.

Second, we evaluate the bounded inverse Laplace transform. The lemma `IsInverseLaplaceBounded'` acts as the pivotal integration by parts step, linking our function f to the `DirichletSin` kernel:

$$\int_0^\infty f(a)e^{-(a-t)\gamma} \frac{T \operatorname{sinc}(T(t-a))}{\pi} da = -f(0)e^{t\gamma}S(-Tt) - \int_0^\infty \frac{d}{da} \left[f(a)e^{-(a-t)\gamma} \right] S(T(a-t)) da$$

Finally, we take the limit as $T \rightarrow \infty$. By combining the aforementioned limits with the Dominated Convergence Theorem (`Tendsto.IntegralDirichletSin.times.integrableFunction'`), we obtain the master theorem. We successfully recover the original function $f(t)$ from its transform:

```
1 theorem IsInverseLaplace (f : ℝ → ℂ) (γ : ℝ) (S : Set ℝ) (hS : ∀ x ∈ S, 0 < x)
2   (h_cont : Continuous f)
3   (h_int : Integrable (fun t ↦ f t * cexp (- (↑γ * ↑t))) volume)
4   (hMeasurable : Measurable f)
5   (h_Laplace_int : ∀ t ∈ S, Integrable ((InverseLaplaceKernelFuncT (RealLaplaceTransform f) t) γ
6     ) volume)
7   (h_diff : Differentiable ℝ f)
8   (h_diff_int : Integrable (fun t ↦ (deriv f t) * cexp (-γ * t))) :
  ∀ (t : S), (inverseLaplaceFunction (RealLaplaceTransform f) γ S h_Laplace_int) t = f t := by
```

By breaking down the Bromwich integral into real-analytic components, we successfully formalize the inverse Laplace transform in Lean 4. While the hypotheses are

more restrictive than Lerch’s theorem, this approach provides a fully verified, foundational tool for mapping frequency-domain solutions back to the time domain without requiring unformalized complex contour geometry.

4.4 Assembling the Inversion Theorem on Lean

In this section, we will focus on the building of the Laplace Inversion Theorem proof, once we proved the Dirichlet’s Integral’s value. Thereby we postpone the proof of the Dirichlet’s Integral’s value to Section 5

4.4.1 Reduction to the Real Domain: The Sinc Transformation

The pivotal step in moving from complex analysis to real analysis is the evaluation of the complex exponential over a symmetric interval $[-T, T]$. We prove that this integral reduces to a real sine-based kernel:

```
1 lemma integral_cexp_Icc_Dirichlet {T t a : ℝ} {hT : 0 ≤ T} :
2   (∫ r in Icc (-T) T, cexp (I * (r : ℂ) * (t - a))) =
3     if h : t - a = 0 then (2 * T : ℂ)
4     else 2 * Real.sin (T * (t - a)) / (t - a)
```

Using this lemma, we apply Fubini’s theorem (`Fubini.lemma`) to swap the order of integration between the Laplace transform and the inverse integral. This leads to the `integrand.simplification` lemma, which expresses the bounded inverse transform as a convolution with a sinc-like kernel:

$$\frac{1}{\pi} \int_0^\infty f(a) e^{-\gamma(a-t)} \frac{\sin(T(t-a))}{t-a} da.$$

4.4.2 Integration by Parts and the Heaviside Transition

To handle the limit as $T \rightarrow \infty$, we cannot rely on a direct pointwise limit of the sinc function. Instead, we perform an integration by parts (IBP) to shift the dependency onto the normalized integral of the sinc function, denoted $S(x)$. We define this function as:

$$S(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^x \frac{\sin u}{u} du$$

In our Lean code, this corresponds to the `DirichletSin` function and its target limit, the `HeavisidePerso` step function:

```
1 noncomputable def DirichletSin : ℝ → ℝ :=
2   fun x ↦ 1/2 + 1/π * ∫ t in (0)..(x), sinc t
3
4 noncomputable def HeavisidePerso (x : ℝ) : ℝ :=
5   if x > 0 then 1 else if x = 0 then 1/2 else 0
6
7 theorem lim_SRx (x : ℝ) :
8   Tendsto (fun R ↦ DirichletSin (R * x)) atTop (ℳ (HeavisidePerso x))
```

The transition from the oscillatory integral to the bounded inversion formula is achieved by applying IBP. By setting $u(a) = f(a)e^{-(a-t)\gamma}$ and recognizing the sinc

kernel as the derivative of the shifted Dirichlet integral, namely $\frac{d}{da}[S(T(a-t))] = \frac{T}{\pi} \text{sinc}(T(a-t))$, we obtain:

$$\int_0^\infty u(a) \frac{d}{da}[S(T(a-t))] da = [u(a)S(T(a-t))]_0^\infty - \int_0^\infty u'(a)S(T(a-t)) da$$

This decomposition is formalized in the following theorem, which introduces $S(T(a-t))$ into the bounded inversion formula:

```

1 theorem IsInverseLaplaceBounded' (f : ℝ → ℂ) (γ T : ℝ) (S : Set ℝ) (...) :
2   ∀ (t : S), ∫ (a : ℝ), f a * cexp (-(↑a - ↑t) * ↑γ) * T * ↑(Real.sinc (T * (↑t - a))) / (↑π) ∂μ
3   _real =
4   -f 0 * cexp (↑t * ↑γ) * ↑(DirichletSin (-T * ↑t)) - ∫ (a : ℝ), deriv (fun u ↦ f u * cexp (-(u -
   t) * γ)) a * DirichletSin (T * (a - t)) ∂μ_real

```

For any $t > 0$, the argument $-Tt$ in the boundary term goes to $-\infty$ as $T \rightarrow \infty$. Since $\lim_{x \rightarrow -\infty} S(x) = 0$, this term vanishes, effectively eliminating the influence of the initial value $f(0)$. In the remaining integral, the term $S(T(a-t))$ acts as a real-analytic selector. As $T \rightarrow \infty$, it converges to 1 for all $a > t$ and to 0 for all $a < t$, effectively isolating the signal on the interval (t, ∞) and allowing for the eventual recovery of $f(t)$.

4.4.3 The Master Theorem

Finally, by combining the convergence of the `DirichletSin` terms and the Dominated Convergence Theorem (`TendstoDirichletIntegral`), we extract the value $f(t)$ from the integral. The final theorem proves that the reconstructed signal is exactly the original function:

```

1 theorem IsInverseLaplace (f : ℝ → ℂ) (γ : ℝ) (S : Set ℝ) (hS : ∀ x ∈ S, 0 < x)
2   (h_cont : Continuous f)
3   (h_int : Integrable (fun t ↦ f t * cexp (-(↑γ * ↑t))) volume)
4   (h_diff : Differentiable ℝ f)
5   (h_diff_int : Integrable (fun t ↦ (deriv f t) * cexp (-γ * t))) :
6   ∀ (t : S), (inverseLaplaceFunction (RealLaplaceTransform f) γ S h_Laplace_int) t = f t

```

The proof uses the uniqueness of limits to bridge the analytical bounds of the truncated Bromwich integral and the pointwise convergence provided by the Dirichlet integral. This formalizes a complete real-analytic path for the Inverse Laplace Transform in Lean 4, ensuring that the inversion is valid for any signal satisfying the standard integrability and differentiability conditions.

5 Formalizing the Dirichlet Integral in Lean 4

To complete the proof of the Inversion Theorem, we must rigorously evaluate the limit of the integral containing the term $\frac{\sin(T(t-a))}{\pi(t-a)}$. This requires proving that:

$$\lim_{T \rightarrow \infty} \int_0^T \frac{\sin(x)}{x} dx = \frac{\pi}{2}$$

In Lean 4, this is non-trivial because the function $\text{sinc}: x \mapsto \frac{\sin x}{x}$ is not Lebesgue-integrable on $(0, \infty)$ (it is not in L^1). Consequently, we cannot use the standard `MeasureTheory.integral` directly over the infinite interval. We must instead formalize it as a limit of interval integrals.

Let us present this proof step by step.

Step 1: Relating the Sinc Integral to its Squared Form

We first relate the sinc function to its square, which is Lebesgue-integrable. We define and prove the integrability of the squared sinc function:

```
1 lemma integrable_sinc_sq : IntegrableOn (fun t ↦ (sinc t)^2) (Ioi 0)
```

This is proven by splitting the domain. On $(0, 1]$, the function is continuous and thus integrable. On $(1, \infty)$, we use a comparison with t^{-2} . This integrability is a prerequisite for using the Dominated Convergence Theorem (DCT) later in the proof.

we now establish a bridge between evaluating the integral of the sinc function and evaluating the integral of its square. This is accomplished through a combination of a change of variables ($u = 2t$) and integration by parts.

```
1 lemma integral_sinc_zero_T (T : ℝ) (hT : T > 0) :
2   (∫ t in 0..T, Real.sinc t) = (∫ t in 0..T/2, (Real.sinc t)^2) + (Real.sinc (T/2))^2 * (T/2)
3
```

This crucial lemma demonstrates that the Dirichlet integral over $[0, T]$ can be expressed exactly as the integral of $\text{sinc}^2(t)$ over half the interval, plus a boundary term. As $T \rightarrow \infty$, the boundary term $\text{sinc}^2(T/2) \cdot (T/2)$ vanishes—a fact formalized separately in the lemma `h_lim_T`. Consequently, evaluating the original limit is safely reduced to evaluating the integral of the squared kernel over $(0, \infty)$.

Step 2: A First Differentiation

To find the exact value of $\pi/2$, we introduce e^{-xt} as a parameter. The strategy, known as "Feynman" technique, consists in differentiating the integral twice with respect to x to successively eliminate the t^2 and t terms in the denominator of $\text{sinc}^2(t)$.

We define the initial integral $I(x)$ and its first derivative $I'(x)$. In Lean, these are represented by the following functions and the corresponding derivation theorem:

```
1 def sinc_sq_times_exp (t : ℝ) : ℝ → ℝ :=
2   fun x ↦ exp (-x * t) * (sinc t)^2
3
4 def integral_sinc_sq_times_exp (x : ℝ) : ℝ :=
5   ∫ t in Ioi 0, sinc_sq_times_exp t x
6
7 def neg_sinc_sq_times_id_exp (t : ℝ) : ℝ → ℝ :=
8   fun x ↦ -(sinc t)^2 * t * exp (-x * t)
9
10 theorem hasDeriv_integral_sinc_sq_times_exp (x : ℝ) (hx : 0 < x) :
11   HasDerivAt (integral_sinc_sq_times_exp) (integral_neg_sinc_sq_times_id_exp x) x
```

The theorem `hasDeriv_integral_sinc_sq_times_exp` proves that the derivative of $I(x)$ is the integral of $-(\text{sinc } t)^2 \cdot t \cdot e^{-xt}$. This operation reduces the power of t in the denominator of the integrand from t^2 to t .

Step 3: Second Differentiation: Reaching the Trigonometric Form

To fully eliminate the denominator, we differentiate a second time. This leads to an integral involving only $\sin^2(t)$ and the exponential:

```
1 def sin_sq_times_exp (t : ℝ) : ℝ → ℝ :=
2   fun x ↦ (sin t)^2 * exp (-x * t)
3
4 theorem hasDeriv_integral_neg_sinc_sq_times_id_exp (x : ℝ) (hx : 0 < x) :
5   HasDerivAt (integral_neg_sinc_sq_times_id_exp) (integral_sin_sq_times_exp x) x
```

The lemma `hasDeriv_integral_neg_sinc_sq_times_id_exp` formalizes this second transition. The t generated by the derivative of the exponential cancels the remaining t in the denominator of the sinc function. This results in the Laplace transform of $\sin^2(t)$, which is computed explicitly:

```
1 theorem compute_deriv_integral_sin_div_times_exp_eps (ε : ℝ) (x : ℝ) (hx : 0 < x) :
2   ∫ t in Ioi ε, (sin t)^2 * exp (-x * t) = rexp (- x * ε) * ((Real.sin (2 * ε) - (x/2) * Real.cos (2 *
   ε)) / (4 + x^2) + 1 / (2 * x))
```

Evaluating this expression as $\varepsilon \rightarrow 0$ gives $I''(x) = \frac{2}{x(x^2+4)}$.

Step 4: Integrating Backwards to the Target Value

Once the second derivative is determined, we perform two successive integrations to recover $I(x)$. The first integration leads to the logarithmic form for $I'(x)$:

```
1 theorem integral_neg_sinc_sq_times_id_exp_eq (x : ℝ) (hx : 0 < x) :
2   integral_neg_sinc_sq_times_id_exp x = 1/4 * log (x^2 / (4 + x^2))
```

Finally, integrating $I'(x)$ provides the analytical expression for the original integral:

```
1 theorem integral_sinc_sq_times_exp_eq (x : ℝ) (hx : 0 < x) :
2   integral_sinc_sq_times_exp x = x/4 * log (x^2/(4+x^2)) + arctan (2/x)
```

Step 5: Final Limit and Convergence

The proof concludes by linking the parametric integral $I(x)$ to the target value $\pi/2$ as the damping factor x vanishes. This transition requires both the analytical limit of the expression found in Step 5 and a measure-theoretic justification to move the limit inside the integral.

To prove that the integral of the damped function converges to the integral of $\sin^2(t)$, we apply the Dominated Convergence Theorem. In Lean, this is formalized using `tendsto_integral_filter_of_dominated_convergence`. The dominating function is $\sin^2(t)$, which was proven integrable in Step 1.

```
1 lemma h_lim_int :
2   Tendsto (fun x ↦ integral_sinc_sq_times_exp x) (N[>] 0) (N (∫ t in Ioi 0, (sin t)^2))
```

This lemma ensures that as $x \rightarrow 0^+$, the term e^{-xt} converges pointwise to 1, allowing us to recover the area under the squared sinc function.

In parallel, we evaluate the limit of the analytical formula $I(x) = \frac{x}{4} \ln(\frac{x^2}{4+x^2}) + \arctan(2/x)$. The logarithmic part is handled by showing that $x \ln x \rightarrow 0$ as $x \rightarrow 0^+$, while the arctangent term reflects the behavior of the function at the origin.

```

1 lemma h_lim_int2 :
2   Tendsto (fun x ↦ (x / 4) * log (x^2 / (4 + x^2)) + arctan (2 / x)) (N[>] 0) (N (π / 2))

```

By the uniqueness of limits, these two results imply that the integral of $\text{sinc}^2(t)$ over the positive real line is exactly $\pi/2$.

```

1 theorem integral_sinc_sq_eq_pi_div_two :
2   ∫ t in Ioi 0, (sinc t)^2 = π / 2

```

The last step uses the identity established in Step 2 to bridge the gap between the squared sinc function and the original sinc function. By taking the limit as $T \rightarrow \infty$, the boundary term $\frac{\sin^2(T/2)}{T/2}$ vanishes, leaving the value of the improper Dirichlet integral.

```

1 theorem integral_dirichlet :
2   Tendsto (fun T ↦ ∫ t in 0..T, sinc t) atTop (N (π / 2))

```

This result confirms that the integral evaluates to the constant required for our Inversion Theorem, formalizing the final link in the real-analytic proof within Lean 4.

References

- [1] Joel L. Schiff, *The Laplace Transform: Theory and Applications*, Undergraduate Texts in Mathematics, Springer, New York, 1999.
- [2] Sangchul Lee, Answer to the question *Proof of inverse Laplace transform* <https://math.stackexchange.com/questions/2926685/proof-of-inverse-laplace-transform>